25901636 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Core Course (CC) - MG2CCRBCA100 - MATHEMATICS FOUNDATION TO COMPUTER SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. What are the main conditions for a graph to have an Euler Path? [U] / [CO1]
2. Every connected graph is a tree. True or False? Justify your answer with an example. [U] / [CO1]
3. Find using Trapezoidal Rule: $\int_0^6 \frac{1}{1+x}\,dx$ [A] / [CO2]
4. Find the Newton-Raphson formula for the function f(x) = $x^3 - x - 3$. [K] / [CO2]
5. What is an objective function in a linear programming problem? [K] / [CO3]
6. Rewrite using artificial variable $4x_1 + 3x_2 \geq 8$ [A] / [CO3]
7. Distinguish between a balanced and unbalanced transportation problem. [A] / [CO4]

8. What is the full form of 'MODI' in transportation problems, and what is the primary purpose of this method? [U] / [CO4]

[2x5 = 10]

Part B

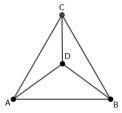
Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. What do you mean by a Connected and a disconnected graph? Construct a Disconnected graph with 5 vertices and 3 components. [A] / [CO1]

10. Produce a spanning tree of the given graph using the spanning tree algorithm [A] / [CO1]



11. Use Newton's Backward formula to find the value of f(x) at x = 7 given the table: [A] / [CO2]

x	2	5	8	11
f(x)	94.8	87.9	81.3	75.1

12. Apply bisection method to find the second approximation to a root of the equation $x^3 - 9x + 1 = 0$. [A] / [CO2]

13. Solve the following LPP using graphical method [A] / [CO3]

Maximize $z = 3x_1 + 4x_2$

Subject to

$x_1 + x_2 \leq 450$

$2x_1 + x_2 \leq 600$

$x_1, x_2 \geq 0$

14. A toy company manufactures dolls and cars. A doll requires 1 hour of assembly and 2 hours of painting. A car requires 3 hours of assembly and 1 hour of painting. The company has 90 hours of assembly time and 80 hours of painting time available. The profit on a doll is Rs. 15, and the profit on a car is Rs. 20. Formulate a linear programming model and solve using simplex method. [A] / [CO3]

15. Find initial feasible solution by Vogels approximation methods. [A] / [CO4]

	D_1	D_2	D_3	Supply
O_1	30	50	45	40
O_2	40	45	38	30
O_3	37	35	39	30
Demand	40	20	40	

16. Obtain the initial basic feasible solution of the following transportation problem using the Least Cost Method. [A] / [CO4]

	Destination			
Origin	D_1	D_2	D_3	*Supply*
O_1	16	20	12	200
O_2	14	8	18	160
O_3	26	24	16	90
Demand	180	120	150	

[6x5 = 30]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **15** marks

17. a) Explain the concept of an Euler cycle and its origin in Graph theory. Mention the conditions necessary for a graph to have an Euler cycle. [A] / [CO1]
b) Give an example of an Euler graph by finding an Euler cycle in it.

18. Find the value of sinØ at Ø = 25º using the Newton - Gregory forward difference formula from the following [K] / [CO2]

Ø	10	20	30	40	50
sinØ	0.1736	0.3420	0.5000	0.6428	0.7660

19. Solve by simplex method [A] / [CO3]
Maximize $z = x_1 - x_2 + 3x_3$
Subject to
$x_1 + x_2 + x_3 \leq 10$
$2x_1 - x_3 \leq 2$
$2x_1 - 2x_2 - 3x_3 \leq 0$
$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$

20. Find the solution for the following transportation problem using the North-West Corner method. [A] / [CO4]

	$D1$	$D2$	$D3$	$D4$	$Supply$
$S1$	19	30	50	10	7
$S2$	70	30	40	60	9
$S3$	40	8	70	20	18
$Demand$	5	8	7	14	

[15x2 = 30]